

RESEARCH ARTICLE

VGG-16, VGG-16 With Random Forest, Resnet50 With SVM, and EfficientNetB0 With XGBoost-Enhancing Bone Fracture Classification in X-Ray Using Deep Learning Models

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ABSTRACT Millions of cases of bone fractures are reported every year, and accuracy in classification is crucial to help with proper management and treatment. The recently developed techniques of Machine Learning, particularly Deep Learning, have been effective in increasing diagnosis precision and efficiency. We utilized a diverse dataset comprising 10 different classes of fracture types captured in X-Ray images. This paper makes a comparison of different machine learning models on classifying bone fractures: VGG-16, VGG-16 with Random Forest, ResNet-50 with Support Vector Machine, and EfficientNetB0 with XGBoost. Model performances were evaluated with respect to parameters of precision, recalls, and F1-scores. According to results, VGG-16 and its variant ensemble with Random Forest outperformed with an accuracy of 0.95 when compared to others on every parameter for different classes of fractures. Results indicate that models based on VGG16 are quite effective for bone fracture classification.

INDEX TERMS Bone fracture, ensemble learning models, fracture classification, machine learning, medical imaging diagnostics, random forest, ResNet-50, VGG-16.

I. INTRODUCTION

Fractures are typically caused by trauma, including falls, road traffic accidents, or sports-related injuries. They can also result from conditions that weaken bones, such as osteoporosis. Osteoporosis, for instance, significantly increases the likelihood of fractures, particularly in older individuals, by reducing bone mass and strength [1]. In most traumatic incidents, the forces exerted on the bone often exceed its shock-absorbing capacity, leading to fractures [2]. Proper

classification of bone fractures is essential for effective management and treatment. Advances in Machine Learning (ML) have significantly enhanced the accuracy and efficiency of fracture diagnosis. Traditionally, diagnosing fractures has relied on manual examinations by radiologists, which are often time-consuming and susceptible to human error. For example, a study demonstrated that deep convolutional neural networks (CNNs) achieved an 83% detection rate for femoral neck fractures [3].

These advancements highlight the potential of ML models in improving diagnostic processes. However, challenges remain. Transfer learning techniques, including VGG-16 and

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ResNet-50, have shown great promise but require further research to identify the most effective configurations. For instance, ResNet-50 has been successfully applied for bone age assessment with reliable results [4]. Nevertheless, the relative performance of these models and the potential benefits of ensemble methods have not been fully explored. This gap presents an opportunity to investigate how different ML algorithms and ensemble techniques can be employed to optimize bone fracture classification.

The objective of this project is to address this gap by conducting a comparative study of various ML models, including VGG-16, ResNet-50, and EfficientNetB0, alongside ensemble learning techniques such as random forest classifiers, support vector machines, and XGBoost. The goal is to identify the most effective combination of models for improving classification accuracy. Previous studies, such as those involving fracture risk categorization, have demonstrated the potential of ensemble techniques when combined with models like VGG-16 and ResNet-50 [5]. This study aims to develop a robust classification framework that offers dependable diagnostic support, potentially surpassing existing methods.

The experiment hypothesizes that combining advanced transfer learning models with ensemble learning techniques will yield superior performance in bone fracture classification tasks. Earlier research has shown that merging ensemble techniques with CNN architectures can result in significant improvements in diagnostic accuracy [6]. This study will test this hypothesis by comparing various model combinations. By leveraging advanced ML models, the research seeks to achieve more accurate bone fracture classification, ultimately enhancing healthcare diagnostics and patient management. The findings are expected to demonstrate that ensembles outperform individual models, paving the way for their wider adoption in clinical settings [7].

II. RELATED WORK

Introduction of deep learning and ensemble methods has resulted in vast improvements in the ML models for classifying bone fractures. Many researchers have looked into different methods and models to increase accuracy and reliability of bone fracture classification. Medical imaging as a classification of bone fractures has seen the adoption of transfer learning models such as VGG-16 and ResNet-50. The study by Hwang et al. used Computed Tomography venography to test the appropriateness of VGG-16 and ResNet-50 models for identifying iliofemoral DVT. They discovered that the VGG-16 model performed slightly better than other traditional ML models like XGBoost [8]. Besides, Eweje et al. created a deep learning model using EfficientNet-B0 which could differentiate between benign and malignant bone lesions to near human radiographer extends [9].

Besides, other studies have shown the advantages of ensemble learning techniques in improving bone fracture classification. Several ensemble methods such as random forests, XGBoost and SVM have been effectively combined

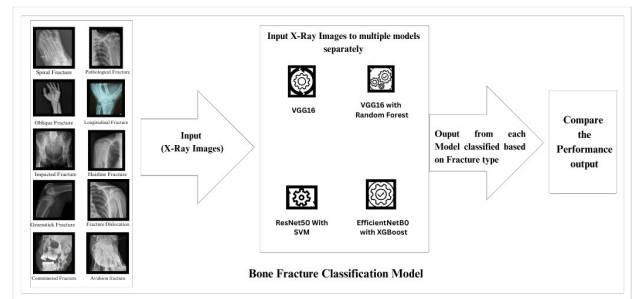


FIGURE 1. Conceptual framework for bone fracture classification leveraging an array of ML methodologies.

with deep learning models to enhance classification accuracy. For example, a study on multi-parametric MRI-based breast cancer classification found out that LightGBM and XGBoost significantly improved its accuracy compared to other models [10]. Another research by Pan et al. employed random forests to classify bone tumors based on conventional radiographic features, in which a high diagnostic performance has been displayed [11]. The most effective ways to classify bone fractures have been shown by comparative studies involving different ML models.

The classification of fractures has benefited from the latest advances in machine learning. 3D CNN and advanced preprocessing techniques have increased the accuracy of models for discriminating between different forms of bone cancers. Consequently, [12] developed a classifier that combines 3D texture features with regional vBMD, which they argue is very effective at identifying vertebral fracture patients. Ensemble learning techniques and transfer learning models have been identified as having enormous capabilities in improving bone fracture classification. These experiments have indicated the possibility of improved diagnostic accuracy and better diagnostic tools by combining these methods, therefore making clinical outcomes better and enhancing patient care.

III. METHODOLOGY

A. OVERVIEW

As illustrated in the figure 1, the workflow for the classification of bone fractures by machine learning models, The dataset was already pre-processed and directly passed into the model, further it involves the removal of top layers from pre-trained deep CNNs—namely, VGG-16, ResNet-50, and EfficientNetB0—for the feature map extraction from bone fracture images. These feature maps are then reshaped into vectors and fed as input to the classifier: VGG-16 with Random Forest, ResNet-50 with SVM, and EfficientNetB0 with XGBoost. In the case of VGG-16, transfer learning with custom dense layers has been used. During training, use the Adam optimizer with categorical cross-entropy loss. The performance of this model will be evaluated using precision, recall, F1-score, and accuracy.

B. DATASET DESCRIPTION

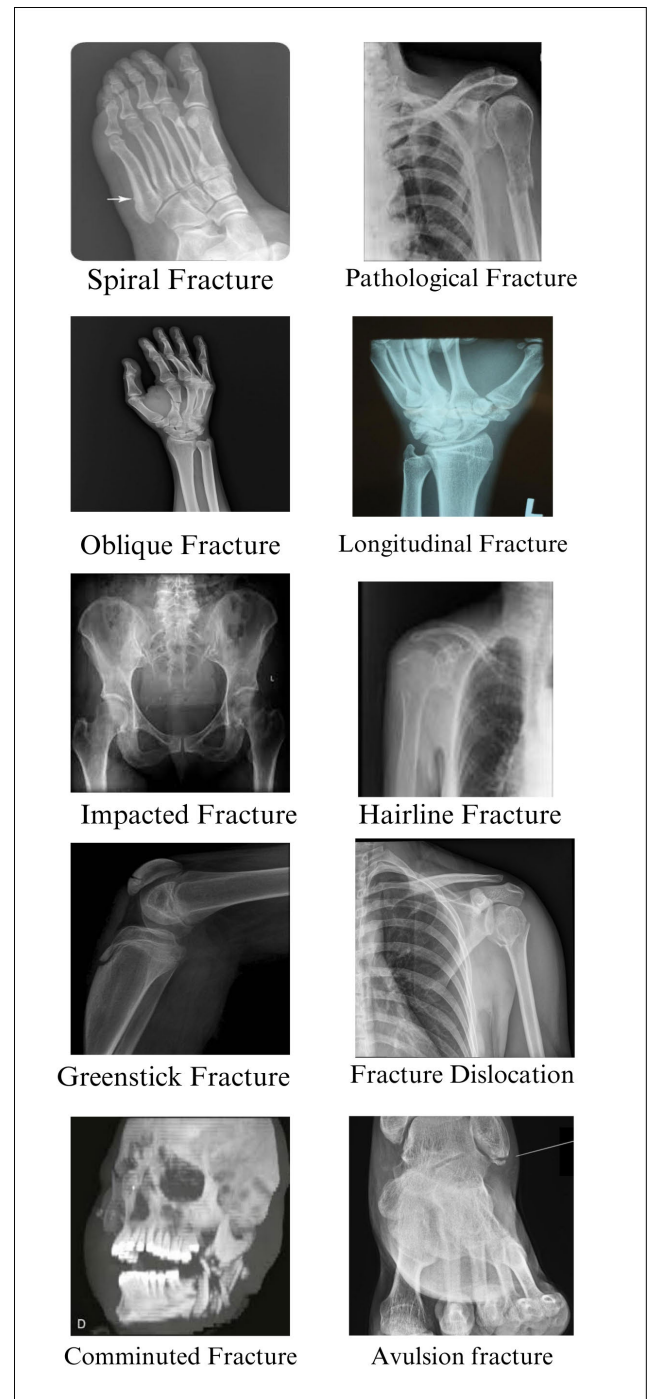
A dataset containing 1,129 images of various types of fractures, including avulsion fractures, comminuted fractures,

TABLE 1. Distribution of images across different fracture classes.

Class (Fracture)	No. of Images(Training)	No.of Images(Testing)
Avulsion	109	14
Comminuted	134	14
Dislocation	137	19
Greenstick	106	16
Hairline	101	10
Impacted	75	9
Longitudinal	68	12
Oblique	69	16
Pathological	116	18
Spiral	74	12
	989	140

fracture-dislocations, greenstick fractures, hairline fractures, impacted fractures, longitudinal fractures, oblique fractures, pathological fractures, and spiral fractures, is key to this aspiration in improving machine learning algorithms that are targeted at automated diagnosis and treatment planning. The distribution of the images as shown in table 1 is, however, somewhat imbalanced, such that classes like ‘longitudinal fractures’ or ‘oblique fractures’ are underrepresented, while proportions are higher for classes like ‘comminuted fractures’ or ‘fracture-dislocations’. The Distributed Dataset Contains 989 images, 140 for testing, and a common split ratio of 0.80 for training and 0.20 for validation. This is a dataset with an input size of $256 \times 256 \times 3$; it comprises gray-scale X-ray images taken from Kaggle as shown in figure 2 [13]. The letter “D” in the image of the Comminuted Fracture likely serves as an orientation marker, indicating “Dextro.” As such, it presents a problem to AI models that detect the subtle differences in fracture patterns and require diverse and labelled datasets to increase diagnostic accuracy with less human error in the clinical setting. Large labelled datasets ensure stability in AI performance for a wide range of patient demographics and various clinical scenarios [14].

There exists a host of problems in collecting and preparation of a quality dataset for fractures. This would need the collaboration of several medical centers to provide a myriad of samples representative of the different causes and severities of the fracture. The consistency in the techniques of X-raying is very important for reliability, just like the uniformity for staining or sample preparation in other medical image datasets [15]. In this way, all samples are accurately presented with regard to the fractures, which will be proactively training the AI models. Another challenge is the labelling of data themselves due to subjective type of interpretation, probably subject to inter-observer variability; different radiologists or researchers may classify a fracture differently. Additional concerns to be brought out on the data collection for bone fractures include patient consent and health records bound to personal information. Conquering this array of challenges can significantly improve the quality of fracture datasets through machine learning technologies, which will have a direct effect on better AI models that can help in diagnostic consistency and improvement in the care of patients.

**FIGURE 2.** Sample dataset images.

C. MODEL ARCHITECTURE

1) **RANDOM FOREST WITH VGG-16(ENSEMBLE LEARNING)**
VGG-16 is a pre-trained ImageNet model for deep CNN used here to extract features by removing its top layers. Thereafter, these images are pre-processed and put through VGG-16 to get the feature maps which are then reshaped into feature vectors. The latter are supplied to a Random Forest classifier, an ensemble learning method that employs many decision trees for improving prediction accuracy and

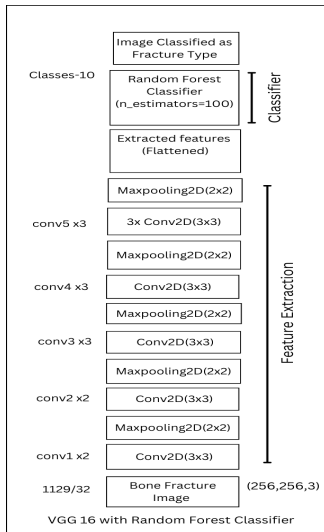


FIGURE 3. VGG16 with random forest classifier (modified) architecture [16].

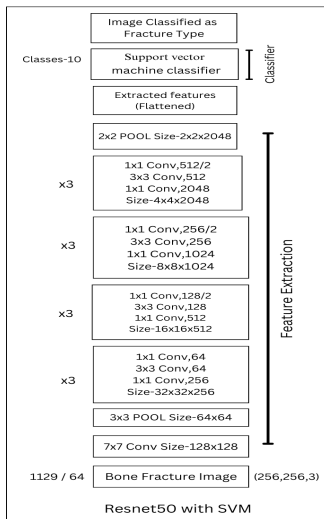


FIGURE 4. Resnet50 with SVM (modified) architecture [17].

controlling over-fitting, as shown in figure 3. This hybrid approach integrates the strengths of Random Forest in classification with the capabilities of VGG-16 in terms of extracting features during deep learning.

2) SVM (SUPPORT VECTOR MACHINE) WITH RESNET-50(ENSEMBLE LEARNING)

ResNet-50 is one of the most common architectures for deep CNN and is known for residual learning that makes training very deep networks easier. As illustrated in Figure 4, we use ResNet-50 without its top layers to get feature maps from input images which can be flattened into feature vectors afterwards. These vectors will be used to train an SVM classifier with a linear kernel. Here we apply SVMs that are good at dealing with high-dimensional spaces for categorizing features obtained from ResNet-50 resulting in a strong ensemble model designed for classification purposes.

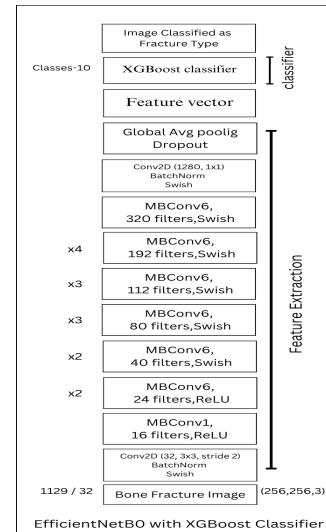


FIGURE 5. EfficientNetB0 with XGBoost classifier (modified) architecture [18].

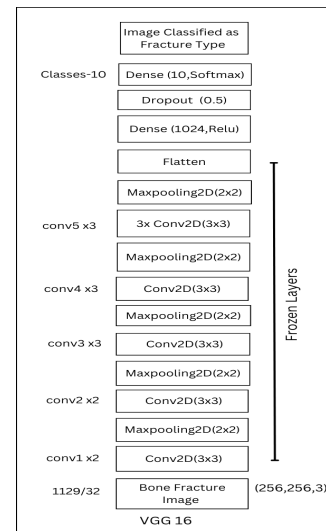


FIGURE 6. VGG16 (modified) architecture [19].

3) EFFICIENTNETB0 WITH XGBOOST (ENSEMBLE LEARNING)

As shown in figure 5, Feature extraction is performed using EfficientNetB0, which is an efficient CNN. Input images are processed by this model to generate feature maps that are then globally pooled and flattened into feature vectors. These vectors are fed into a classifier called XGBoost, a well-optimized gradient boosting algorithm, known for its efficiency and accuracy. XGBoost is excellent at handling massive datasets and complicated models; thus, it can take advantage of the deep features generated by EfficientNetB0 to create a powerful ensemble model suitable for classification tasks.

4) VGG-16 (TRANSFER LEARNING)

In this method of transfer learning, the VGG-16 model is utilized as the starting point such that its layers remain frozen with their learned weights retained. As it is shown in

Figure 6, on top of the model, custom layers have been added with one flattening layer and two dense layers, all of them having ReLU as an activation function. Additionally, one dropout layer was also placed to regularize the system. The above mentioned had been added to adapt model predictions according to the paper's specific needs. The output layer employs SoftMax activation for multi-class prediction. The whole model is compiled using Adam optimizer and trained using a new data set by performing feature extraction through VGG-16 while fine-tuning newly added ones to suit particular types of classification problems. These training steps also included monitoring and enhancing performance with Tensor Board or checkpoints.

D. MODEL TRAINING SETUP

While fine-tuning a VGG-16 model for a classification problem with 10 output classes, the last layer should apply the SoftMax activation function as it is often done in multi-class classifications. The Adam optimizer is used to compile the model at a learning rate of 0.0005 that offers a trade-off between stability and speed. This categorical cross-entropy loss function is the most suitable for solving multi-class classification tasks. During training, the model is fit on xtrain and ytrain over 20 epochs with a batch size of 32. Shuffling data ensures that the trained models generalize well while validation data were used for checking its performance against new data not seen during the training phase. Then, we saved our models using Model Checkpoint callbacks and logged our training progress via Tensor Board callback during the training process.) These settings guarantee robust training procedures that can be rescued using checkpoints or extensively observed to measure development in real-time. CCE-Categorical Cross-entropy. [20]

$$\text{CCE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C (y_{i,c} \times \log(p_{i,c})) \quad (1)$$

N: Number of samples, C: Number of classes, $y(i,c)$ True binary indicator (0 or 1) if class label c is the correct classification for sample i , $p(i,c)$ Predicted probability that sample i belongs to class c .

E. EVALUATION MATRIX

For complete evaluation of classification models, several metrics are taken into consideration. The formulas used to calculate are mentioned in figure 7, Precision gives the proportion of correct positive predictions, while recall (sensitivity) includes all actual positives found by the model. The F1-score is a balance between precision and recall, into a single statistic, taking into account false positives and negatives. Support refers to the number of instances for every class, and accuracy indicates the proportion of correct predictions in all classes [21]. Macro average deals with all classes equally in the calculation of metrics, and weighted average handles class imbalances. Specificity speaks about the true negatives, hence reducing the false

Metric	Formulae
Precision	$\frac{TP}{TP + FP}$
Recall	$\frac{TP}{TP + FN}$
F1-Score	$2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
Accuracy	$\frac{TP + TN}{TP + FP + TN + FN}$
Support	$TP + FN$
Sensitivity	$\frac{TP}{TP + FN}$
Specificity	$\frac{TN}{TN + FP}$
True Positive Rate (TPR)	$\frac{TP}{TP + FN}$
False Positive Rate (FPR)	$\frac{FP}{FP + TN}$
Macro avg - Precision	$\frac{\text{Precision1} + \text{Precision2} + \dots + \text{Precision N}}{N}$
Macro avg - Recall	$\frac{\text{Recall1} + \text{Recall2} + \dots + \text{Recall N}}{N}$
Macro avg - F1 Score	$\frac{\text{F1 Score1} + \text{F1 Score2} + \dots + \text{F1 Score N}}{N}$
Weighted avg - Precision	$\frac{\sum (\text{Precision}(i) \cdot \text{Support}(i))}{\sum \text{Support}(i)}$
Weighted avg - Recall	$\frac{\sum (\text{Recall}(i) \cdot \text{Support}(i))}{\sum \text{Support}(i)}$
Weighted avg - F1 Score	$\frac{\sum (\text{F1 - Score}(i) \cdot \text{Support}(i))}{\sum \text{Support}(i)}$

FIGURE 7. Metrics table.

positives [22]. These measures give different views on the performance of classification and, therefore, make an exhaustive understanding of the model's strengths and weaknesses.

Accuracy, sensitivity, and specificity are very important parameters primarily used in the evaluation of classification systems. Accuracy is a simple measure for exactness, it calculates the rate of correct prediction for all classes. Sensitivity/recall is key in medical diagnosis due to its tendency to find the true positives. High specificity then emphasizes a correct identification of the true negatives, which is important for a low false positive rate. Therefore, these metrics would be able to provide a fair and comprehensive assessment and become important standards during model evaluation benchmarks.

IV. EXPERIMENTAL RESULTS

The table 2 contains the precision, recall, and F1-scores for various classes of fractures against four models: VGG-16, VGG-16 with random forest, ResNet50 with support vector machine, and EfficientNetB0 with XGBoost. All classes of fractures returned very high scores for VGG-16 and VGG-16 with random forest; all were about 0.95 in macro and weighted averages, making these very strong classifiers for fractures. Although ResNet-50 with Support Vector Machine works quite well, it clearly lags behind VGG-16 variants with a macro and weighted average of 0.93. In sharp contrast, EfficientNetB0 with XGBoost substantially underperformed, doing not just very bad in the queries related to recall and F1-measure across most types of fractures but showing only around 0.42-0.43 by macro and weighted averages. From this, one can very clearly understand the

TABLE 2. Experimental results.

Models	VGG-16			VGG-16 with Random Forest			ResNet-50 With SVM			EfficientNetB0 with XGBoost		
Class (Fracture)	precision	Recall	F1-Score	precision	Recall	F1-score	precision	Recall	F1-score	precision	Recall	F1-score
Avulsion	1.00	1.00	1.00	0.92	1.00	0.96	0.93	0.92	0.93	0.38	0.32	0.35
Comminuted	1.00	1.00	1.00	1.00	0.87	0.93	0.94	0.90	0.92	0.44	0.60	0.51
Dislocation	0.89	1.00	0.94	0.89	1.00	0.94	0.91	0.96	0.93	0.46	0.55	0.50
Greenstick	1.00	0.80	0.89	0.90	0.90	0.90	0.92	0.94	0.93	0.41	0.56	0.47
Hairline	0.91	1.00	0.95	0.91	1.00	0.95	0.93	0.95	0.94	0.39	0.36	0.37
Impacted	0.88	0.88	0.88	1.00	1.00	1.00	0.92	0.92	0.92	0.43	0.20	0.27
Longitudinal	0.92	0.92	0.92	1.00	0.92	0.96	0.94	0.95	0.94	0.30	0.20	0.24
Oblique	1.00	0.90	0.95	1.00	0.90	0.95	0.90	0.93	0.91	0.57	0.21	0.31
Pathological	1.00	0.92	0.96	1.00	0.92	0.96	0.95	0.90	0.92	0.55	0.67	0.60
Spiral	0.88	1.00	0.93	0.88	1.00	0.93	0.94	0.93	0.94	0.30	0.25	0.27
Macro avg	0.95	0.94	0.94	0.95	0.95	0.95	0.93	0.93	0.93	0.42	0.39	0.39
Weighted avg	0.95	0.95	0.95	0.95	0.95	0.95	0.93	0.93	0.93	0.43	0.43	0.42

better performance by VGG16-based models in classifying different types of bone fractures than the EfficientNetB0 with XGBoost model. Table 2 provides a comprehensive overview of model performance metrics.

V. RESULTS AND DISCUSSION

In the fracture type classification, the best models were VGG-16 and its ensemble counterpart: VGG-16 with Random Forest. This model could generally achieve accuracy as high as 95%. High sensitivity and specificity values in all classes of fractures establish that these models as very reliable in differentiating different kinds of fractures. The paper establishes a 100% sensitivity and specificity for avulsion, dislocation, and even hairline fractures. Specifically, what sets this model apart is its deep 16-layer architectural design, which makes it efficient for the purpose of feature extraction and representation learning. An in-depth exploration allows it to learn further detailed and complex patterns about the images themselves, which enables delivery of outstanding performances in classification. The random forest component of the ensemble model prunes the outcome by conglomerating output from not one, but several decision trees into classification. This not only avoids overfitting but also boosts generalization by invoking all the strengths of various trees to normalize the desired features in input data.

On the other hand, it was found that EfficientNetB0 with XGBoost performed very poorly with an overall accuracy of 41%. Worst of all sensitivities was the ‘Fracture’ class, with 6% for impacted fractures, along with generally poor specificity. This underperformance could be due to the fact that EfficientNetB0 is probably just one strong model—indeed known not to work very well with relatively smaller images and grayscale images as compared with models like VGG16. This could be an indication that EfficientNetB0’s architecture was focused essentially on model scaling and did not aim at revealing a set of descriptive parameters related to fine-grained features that would characterize this scenario. Second, although potent in some scenarios, such extractive featurization did not synergize well with the peculiarities of the fracture dataset when combined in the

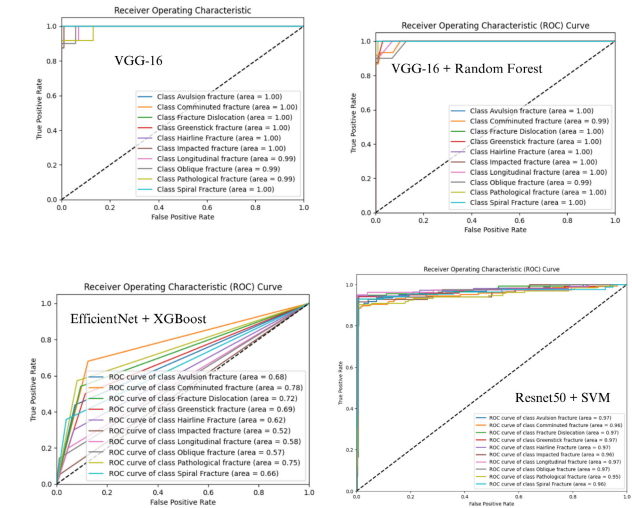


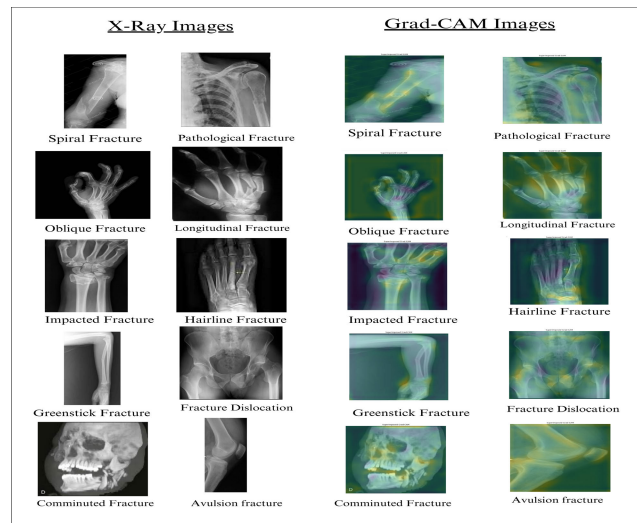
FIGURE 8. ROC curves of all 4 models.

context of XGBoost. This might be because of the fact that the gradient-boosting algorithm present in XGBoost is being deluged by the convolutional features from EfficientNetB0, leading to suboptimal feature choice and poor classification. It thus failed to extract the subtlety of the data, and since then, with poor generalization, it showed low sensitivity over almost all fracture classes. Table 3 provides a comprehensive overview of model performance metrics.

The The figure 8 shows the Receiver Operating Characteristic curves for four different models used in bone fracture classification: VGG-16 with Random Forest, VGG-16, EfficientNetBo + XGBoost, and Resnet50 with SVM. Each subplot represents the ROC curve for each model, showing the variation of the true positive rate against the false positive rate for ten different classes belonging to fractures. The curves with corresponding AUC values quantize the performance of each model with respect to class differentiation. The VGG-16 and VGG-16 with Random Forest models achieve very high AUC values, values very close to 1.00 over all classes, thus reflecting their high accuracy

TABLE 3. Testing results.

Class (Fracture)	VGG-16		VGG-16 + Random Forest		ResNet-50 + SVM		EfficientNetB0 + XGBoost	
	sensitivity	specificity	sensitivity	specificity	sensitivity	specificity	sensitivity	specificity
Avulsion	1.00	1.00	1.00	1.00	0.92	0.99	0.44	0.85
Comminuted	1.00	1.00	0.87	0.99	0.90	0.99	0.68	0.86
Dislocation	1.00	1.00	1.00	1.00	0.96	0.98	0.54	0.86
Greenstick	0.80	0.98	0.90	0.99	0.94	0.98	0.50	0.88
Hairline	1.00	1.00	1.00	1.00	0.95	0.99	0.30	0.88
Impacted	0.88	0.99	1.00	1.00	0.92	0.99	0.06	0.91
Longitudinal	0.92	0.99	0.92	1.00	0.95	0.99	0.20	0.95
Oblique	0.90	0.99	0.90	0.99	0.93	0.99	0.14	0.85
Pathological	0.92	0.99	0.92	0.99	0.90	0.98	0.57	0.88
Spiral	1.00	1.00	1.00	1.00	0.93	0.99	0.36	0.89
Accuracy	0.95		0.95		0.93		0.41	

**FIGURE 9.** X-Ray images and Grad-CAM images.

in classification. In sharp contrast, the EfficientNetB0 with XGBoost model has low AUC values, indicating that it is very poor at differentiating between types of fractures. The Random Forest with SVM performance is quite robust with high AUC values, if not as perfect as the VGG-16 models.

The figure 9 shows a comparative study of original X-ray images against their Grad-CAM visualizations in different kinds of fractures. On the extreme left column are the images that show different types of fractures on X-ray images, such as Spiral, Pathological, Oblique, Longitudinal, Impacted, Hairline, Greenstick, Fracture Dislocation, Comminuted, and Avulsion. Grad-CAM images in the right column will highlight regions of the X-rays where the deep learning model puts more attention when predicting the type of fracture. Grad-CAM visualizations will vividly show the critical areas in the X-rays that more contributed to the decision-making process in the prediction of the model. It clearly shows that the model is correct in focusing on the clinically relevant features of X-ray images and therefore validates its potential as a reliable tool for automated fracture diagnosis.

Although, VGG-16 (40.11 GFLOPs) yields better performance on the task of classification compared to

TABLE 4. Models parameters and GFLOPs.

Models	Model Parameters	GFLOPs
VGG-16	14714688	40.114847744
ResNet-50	23587712	10.12383296
EfficientNetB0	4049571	0.800722455

ResNet-50 and EfficientNetB0. That means VGG-16, while resource-intensive, is more relevant to the representation of relevant features within the bone fracture dataset. While the other models are much more efficient concerning GFLOPs according to table 4, the trade-off among model complexity, computational demands, and accuracy makes VGG-16 a more effective choice regarding this particular problem.

VI. REAL WORLD APPLICATION

Applications of real data from models, such as VGG-16, VGG-16 with Random Forest, ResNet-50 with SVM, and EfficientNetB0 with XGBoost, can significantly improve the accuracy and speed of diagnosis related to bone fractures. Such models can be linked to any hospital imaging system; thus, radiologists would find it much easier to identify several types of fractures quickly and accurately from the given images of X-rays. For instance, when a radiologist uploads an X-ray image, it will classify the type of fracture. It yields immediate feedback and hence aids in making swift decisions. This might be pretty helpful in emergency cases where time is a crucial factor. In addition, the high sensitivity of the models ensures that even very minute fractures are not missed, hence reducing misdiagnosis and enhancing the outcome for the patient. There are some practical considerations while deploying these models in real-world scenarios. First and foremost, these are computational resources for training and inference. Models like VGG-16 and ResNet-50 with SVM may have huge computational and memory requirements hence, powerful hardware or cloud-based solutions would be required. These models have to be very easily integrated into current workflows so that a radiologist or health professional could just pick up the technology and work with it without much training. Data privacy and security go very much in line with handling

patients' sensitive information. Ensuring compliance with regulations such as HIPAA is crucial [23]. Finally, continuous model monitoring and updating are necessary to keep this high performance since real-world data can vary and evolve over time. It requires regular retraining and validation to ensure that the models remain accurate and reliable. One of the study highlights the clinical importance of AI in medical diagnostics, its potential to make diagnoses more accurate, reduce errors in diagnosis, and optimize the outcome of treatments. About 73.68% of healthcare practitioners believe that AI is effective in reducing diagnostic errors and is the future of medical diagnostics. Moreover, 65.55% of respondents agreed that AI can dramatically improve diagnostic accuracy, while the rest also believe in its positive effects. By promising early disease detection and enhancement of precision, AI is expected to change the face of patient care with ethical issues, unequal access, and data security challenges [24].

VII. CONCLUSION

Comparative studies of the machine learning model with respect to bone fracture classification reveals better performances of VGG-16 and its ensemble variant with random forest. Besides, the accuracy, sensitivity, and specificity of these models are quite well in line, making them very reliable models for differentiating different types of fractures. At the same time, EfficientNetB0 along with XGBoost portrays quite a poor trend and hence proves to be less efficient for this task. A good estimate of the improved performance of VGG-16 could be due to the deep architectural design that makes it efficient in feature extraction and representation learning. Obviously, execution time depends on many parameters such as image size. These models thus could be integrated into hospital imaging systems to aid radiologists in the fast identification of a fracture and hence improve patient outcomes. However, the work is limited by the variation in the data distribution across fracture classes and the inherent variability of fracture patterns, which could impact generalizability. Further, model optimization and improved techniques of ensemble use in medical imaging diagnostics will greatly improve the accuracy and efficiency of future studies.

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To ensure academic honesty, this manuscript was prepared utilizing AI-driven techniques that cite the proper articles, increase understanding through language clarity, and provide extensive literature analysis. Citations were managed using (<https://www.citethisforme.com>). They used Grammarly and Paperpal to check for typos, punctuation issues, and grammatical flaws in the sentence. Also, they offered style recommendations that improved the manuscript's readability and professionalism. Simultaneously preserving the original context and content, ChatGPT/Quillbot's comprehension power brings complex thoughts to a more concise expression. With the aid of Scopus AI, a generative AI (GenAI) driven search tool that is both clever and intuitive, they were able

to comprehend and deepen their discoveries at a rate and with details never seen before. Since Scholarcy aggregated relevant academic publications and important discoveries, it aided in bringing together existing research and identified research gaps, which sped up the process. They utilized Turnitin software to detect instances of similarity.

DECLARATION OF COMPETING INTERESTS

Authors declare that they have no conflict of interest. All authors have agreed to the published version of the manuscript.

AUTHORS CONTRIBUTION

Spoorthy Torne: methodology, formal analysis, investigation, writing, review and editing; Dasharathraj K. Shetty: conceptualisation, methodology, visualisation and supervision; Krishnamoorthi Makkithaya: supervision and review; Prasiddh Hedge: validation and review; Manu Sudhi: validation and review; Phani Kumar Pullela: data curation; Tamil Eniyan T: methodology, project administration; Ritesh Kamath: writing, project administration; Staissy Salu: writing, review and editing; Pranav Bhat: data curation; Girisha S: Project administration, supervision and edit; Priya P S: Supervision and review.

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